JEE-Main-01-02-2024 (Memory Based) [MORNING SHIFT]

Physics

Question: The dimensions of angular impulse is equal to:
Options:
(a) $[ML^2T^{-1}]$
(b) $[ML^2T]$
(c) $[ML^2T^2]$
(d) $[MLT^{-1}]$
Answer: (a)

Question: A vernier caliper has 10 main scale divisions coinciding with 11 vernier scale division equals 5 mm. The least count of the device is:
Options:
(a) $\frac{1}{2} mm$
(b) $\frac{5}{12} mm$
(c) $\frac{5}{11} mm$
(d) 0.3 mm
Answer: (c)

Question: On increasing temperature, the elasticity of a material:
Options:
(a) Increases
(b) Decreases
(c) Remains constant
(d) May increase or decrease
Answer: (b)

Question: Determine the lowest energy of photon emitted in Balmer Series of hydrogen atom.
Options:
(a) 10.02 eV
(b) 1.88 eV
(c) 1.65 eV
(d) 2.02 eV
Answer: (b)

Question: De Broglie wavelength of proton = λ and that of an α particle is 2λ. The ratio of velocity to proton to that of α particle is:
Options:
(a) 8

(b) ⅛
(c) 4
(d) ¼
Answer: (b)

Question: 2 moles of a monatomic gas and 6 moles of a diatomic gas are mixed. Molar specific heat for constant volume of the mixture shall be___
(R is the universal gas constant)
Options:
(a) 1.75 R
(b) 2.25 R
(c) 2.75 R
(d) 2.50 R
Answer: (b)

Question: A gas undergoes a thermodynamics process from state (P_1, V_1, T_1) to state (P_2, V_2, T_2). For the given process $PV^{3/2}$ = constant, find the work done by the gas
Options:
(a) $\frac{P_2V_2 - P_1V_1}{2}$
(b) $\frac{P_1V_1 - P_2V_2}{2}$
(c) $\frac{3(P_1V_1 - P_2V_2)}{2}$
(d) $2(P_1V_1 - P_2V_2)$
Answer: (d)

Question: Two particles each of mass 2 kg are places as shown in x - y plane. If the distance of centre of mass from origin is $\frac{4\sqrt{2}}{x}$, find x:
Options:
Answer: ($x = 2$)

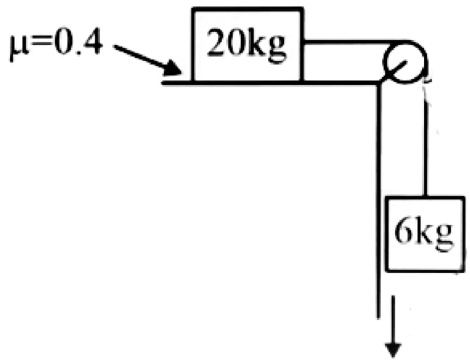
Question: A bullet of mass 10^{-2} kg and velocity 200 m/s gets embedded inside the bob of mass 1 kg of a simple pendulum. The max. Height that the system rises by is___ cm.
Options:
Answer: (20)

Question: The length of a seconds pendulum if it is placed at height 2R from the surface of the earth
(R : radius of earth) is $\frac{10}{x\pi^2}m$. Find x.
Options:
Answer: (9)

Question: Find percentage change in capacitance if potential difference across it has been changed from V to 2V.
Options:

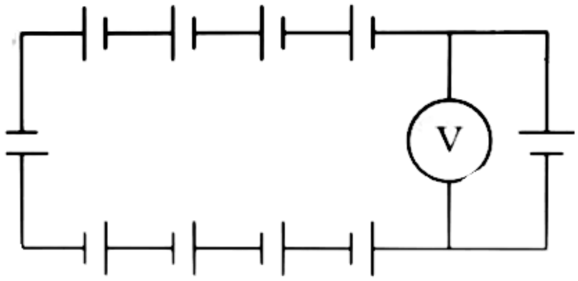
(a) 0
(b) 1.5
(c) 3
(d) 6
Answer: (a)

Question: Find acceleration of the system if an external force of 60 N is applied on 6 kg block as shown.



Options:
(a) $\frac{20}{13} m/s^2$
(b) $\frac{20}{12} m/s^2$
(c) $\frac{20}{14} m/s^2$
(d) $\frac{20}{16} m/s^2$
Answer: (a)

Question: All batteries are identical (5V, 0.2 Ω) and connected as shown in figure find the reading of voltmeter.



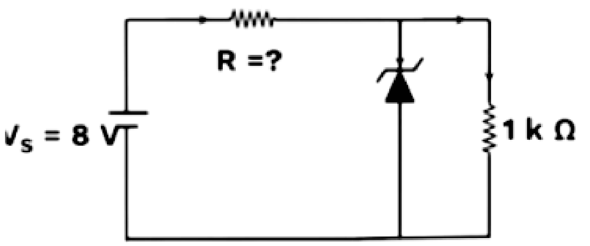
Options:
(a) -20 V
(b) -10 V
(c) 10 V
(d) 0 V
Answer: (d)

Question: Find velocity when acceleration is 0
x = - 3t^3 + 18t^2 + 16t
Options:

(a) 46 m/s
(b) 52 m/s
(c) 25 m/s
(d) 100 m/s
Answer: (b)

Question: In a series LCR circuit if capacitance is changed from C to 4C. How should the inductance be changed so that circuit has same resonant frequency as before.
Options:
(a) Reduced by L/4
(b) Reduced by 3L/4
(c) Reduced by L/2
(d) Reduced by L
Answer: (b)

Question: Breakdown voltage of Zener diode is 5V. Lower consumed across zener is 20 mW



Options:
(a) 5 kΩ
(b) 3/7 kΩ
(c) 10 kΩ
(d) 5/7 kΩ
Answer: (b)

Question: $I = 3t^2 + 4t^3$. Determine charge passing through in
$t = 1$ to $t = 2$ sec.
Options:
(a) 20
(b) 21
(c) 22
(d) 23
Answer: (c)

Question: Find magnetic field at the centre of a hexagonal loop of total length 4π carrying current of $4\sqrt{3}\pi$.
Options:
(a) 72×10^{-7}
(b) 60×10^{-7}
(c) 72×10^{-5}
(d) 60×10^{-6}
Answer: (a)

Question: Two identical charged masses (density = 1.5 g/cc) are suspended from two strings from a common point are is equilibrium in air at angle θ with vertical. If setup is immersed in water and angle remains same find K of medium.

Options:

(a) 1

(b) 2

(c) 3

(d) 4

Answer: (c)

Question: Radius of a nucleus is 4.8 fermi and mass no. is 64. Find atomic mass of nucleus of radius 4 fermi.

Options:

Answer: (27)

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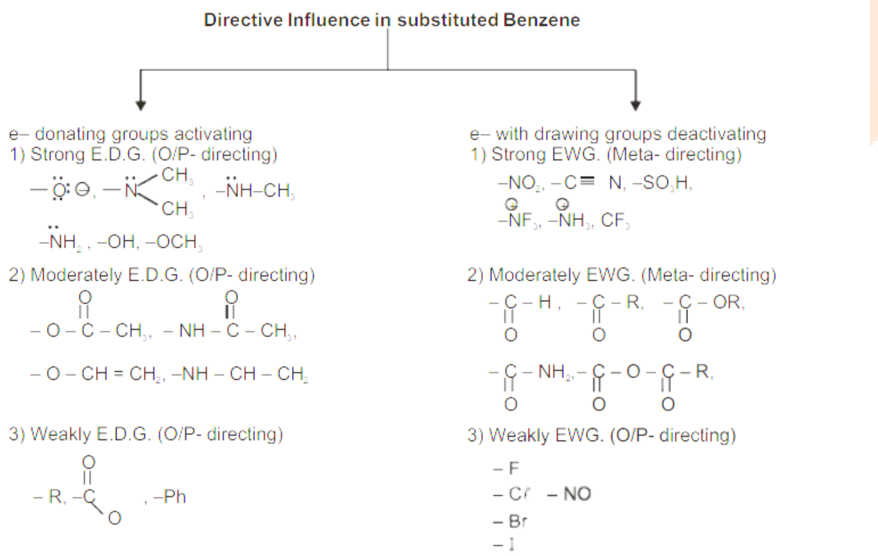
Chemistry

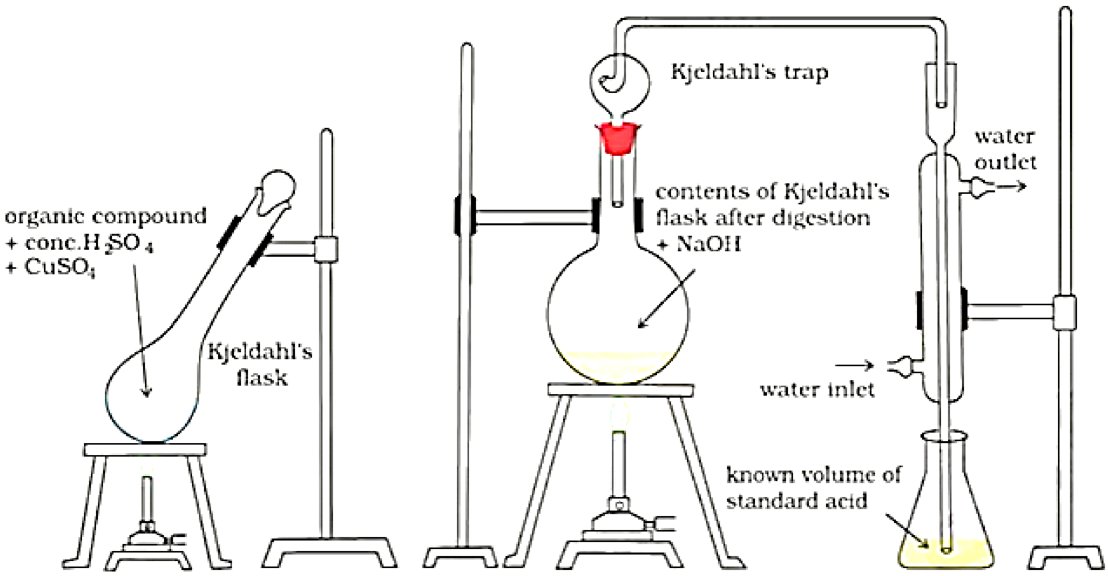
Question: Find out total possible optical isomers of 2-chlorobutane.
Options:
(a) 2
(b) 3
(c) 4
(d) 6
Answer: (a)
Solution: The number of optical isomers possible for a compound is r where n = number of asymetric carbon atoms.
As $2^n = 1$ for 2-ehlorobutane,
$2^n = 2^1 = 2$.
Hence, it has two optical isomers.

Question: The total number of deactivating groups among the following is : -CN, - NH-CO-CH_3, -CO-CH_3, -NH-CH_3
Options:
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (b)
Solution:



Question: In Kjeldahl's estimation of nitrogen, $CuSO_4$ acts are :
Options:
(a) Oxidising agent

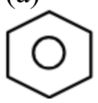
(b) Reducing agent
(c) Catalyst
(d) Reagent
Answer: (c)
Solution:



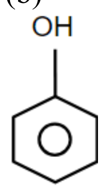
digestion (heating in Kjeldahl's flask)   Kjeldahl's distillation

Taken and that left after the reaction gives the amount of acid reacted with ammonia.
Organic compound + $H_2SO_4 \rightarrow (NH_4)_2SO_4 \xrightarrow{2NaOH} Na_2SO_4 + 2NH_3 + 2H_2O$
$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

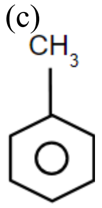
Question: Which of the following is most likely attacked by electroph
Options:
(a)



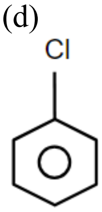
(b)



(c)



(d)



Answer: (c)

Question: Statement I : S_8 disproportionates into $H_2S_2O_3$ and S_2^- in alkaline medium
Statement II : ClO^-_4 undergoes disproportionation in acidic medium
Options:
(a) Statement I is correct but Statement II is incorrect
(b) Statement I is incorrect but Statement II is correct
(c) Both Statement I and Statement II are correct
(d) Both Statement I and Statement II are incorrect
Answer: (c)
Solution:

Phosphorous, sulphur and chlorine undergo disproportionation in the alkaline medium as shown below :

$$\overset{0}{P_4}(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow \overset{-3}{PH_3}(g) + 3H_2\overset{+1}{P}O_2^-(aq) \quad (8.46)$$

$$\overset{0}{S_8}(s) + 12\ OH^-(aq) \rightarrow 4\overset{-2}{S^{2-}}(aq) + 2\overset{+2}{S_2}O_3^{2-}(aq) + 6H_2O(l) \quad (8.47)$$

$$\overset{0}{Cl_2}(g) + 2\ OH^-(aq) \rightarrow \overset{+1}{Cl}O^-(aq) + \overset{-1}{Cl^-}(aq) + H_2O(l) \quad (8.48)$$

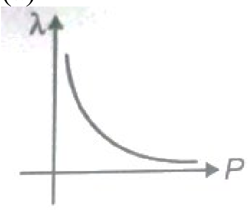
Question: Match the following and select the correct option.

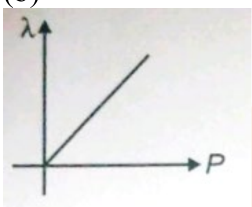
List I	List II
(a) $[Cr(H_2O)_6]^{3+}$	(i) $t_{2g}^2\ eg^0$
(b) $[Fe(H_2O)_6]^{3+}$	(ii) $t_{2g}^3\ eg^0$
(c) $[Ni(H_2O)_6]^{2+}$	(iii) $t_{2g}^3\ eg^2$
(d) $[V(H_2O)_6]^{3+}$	(iv) $t_{2g}^6 eg^2$

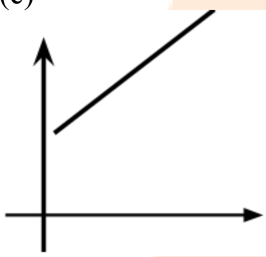
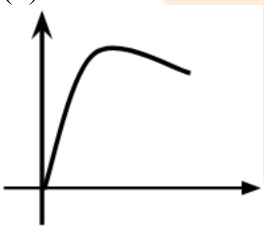
Options:
(a) a-ii, b - iii, c - iv, d - i
(b) a-iii, b - iv, c - i, d - ii
(c) a-iv, b - ii, c - iii, d - i
(d) a-ii, b - iv, c - i, d - iii
Answer: (a)

Question: Statement I: PH_3 will have lower boiling point than NH_3.
Statement II: There are strong van der Waals forces in NH_3 and strong hydrogen bonding in PH_3
Options:
(a) Both Statement I and Statement II are correct
(b) Both Statement I and Statement II are incorrect
(c) Statement I is correct, but Statement II is incorrect
(d) Statement I is incorrect, but Statement II is correct
Answer: (c)

Solution: NH_3 intermolecular hydrogen bonding leads to molecular association so large amount of energy is required to break these hydrogen bonds whereas in PH_3 there is no hydrogen bonding. Hence NH_3 has high boiling point than PH_3

Question: Which of the following is the correct plot between $\boldsymbol{\lambda}$ (de-Broglie wavelength) and p (momentum) ?
Options:
(a)


(b)


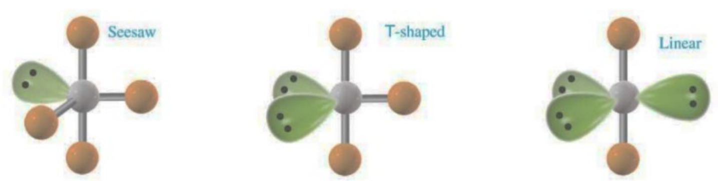
(c)

(d)

Answer: (a)
Solution: Graph between λ and p is a rectangular hyperbola

Question: What is the pH of $CH_3COO^-NH_4^+$? (At 25ºC)
Given: K_a of $CH_3COOH = 1.8 \times 10^{-5}$, Kb of NH_4 OH = 1.8×10^{5}
Options:
(a) 7
(b) 9
(c) 8.9
(d) 7.8
Answer: (a)

Solution: $pH = 7 + \frac{1}{2}(pK_a - pK_b)$

Question: Which of the following is correct for adiabatic free expansion against vacuum ?
Options:
(a) $q = 0$, $\Delta U = 0$, $w = 0$
(b) $q \neq 0$, $w = 0$, $\Delta U = 0$
(c) $q = 0$, $\Delta U \neq 0$, $w = 0$
(d) $q = 0$, $\Delta U \neq 0$, $w \neq 0$,
Answer: (a)
Solution: Free expansion of an ideal gases under adiabatic condition is $q = 0$, $\Delta T = 0$ and $w = 0$.

Question: Which of the following have trigonal bipyramidal shape ?
PF_5 , PBr_5, $[PtCl_4]^-$, SF_6, BF_3, BrF_5, PCl_5, $[Fe(CO)_5]$
Options:
(a) PF_5 , PBr_5, PCl_5, $[Fe(CO)_5]$ only
(b) PF_5 , PBr_5, PCl_5, BrF_5 only
(c) PF_5 , PCl_5, $[Fe(CO)_5]$ only
(d) PF_5 , PBr_5, BrF_5 , PCl_5, $[Fe(CO)_5]$ only
Answer: (a)
Solution:



Question: Complementary stand of DNA ATGCTTCA is :
Options:
(a) TACGAAGA
(b) TACGAAGT
(c) TAGCAACA
(d) TAGCTACT
Answer: (b)
Solution: A always pairs with T with two hydrogen bonds and G always pairs with C with three hydrogen bonds.

Question: We are given with 3 NaCl samples and their Van't Hoff factors

Sample	Van't Hoff factor
Sample - 1 (0.1 M)	i_1
Sample - 2 (0.01 M)	i_2
Sample - 3 (0.001 M)	i_3

Options:
(a) $I_1 = i_2 = i_3$
(b) $I_1 > i_2 > i_3$
(c) $I_3 > i_2 > i_1$
(d) $I_1 > i_3 > i_2$

Answer: (a)
Solution: $I_1 = i_2 = i_3$

Question: How many oxides are amphoteric in nature ?
SnO_2, PbO_2, SiO_2, P_2O_5, Al_2O_3 CO_2 CO, NO, N_2O
Answer: 3

Question: We are given with following cell reaction :
$2H^+ + 2e^- \rightarrow H_2$
$P_{H2} = 2$ atm
$[H^+] = 1$ M
(2.30RT/F=0.06)
If E_{cell} of the reaction is given by - x ✕ 10^{-3} V. Find out x.
Answer: 9
Solution:
Anode: H_2 (1 atm) → $2H^\oplus$ (x M) + $2e^-$
Cathode: $2H^\oplus$ (1 M) + $2e^-$ → H_2, (1 atm)

$$= E^\circ_{H^+/\frac{1}{2}H_2} + \frac{0.059}{n} \log \frac{[H^+]}{(p_{H_2})^{1/2}}$$

JEE-Main-01-02-2024 (Memory Based) [MORNING SHIFT]

Mathematics

Question: Five people are distributed in four identical rooms. A room can also contain zero people. Find the number of ways to distribute them.
Options:
(a) 47
(b) 53
(c) 43
(d) 51
Answer: (d)
Solution:

Question: $3, a, b, c$ are in AP, $3, a-1, b+1, c+9$ are in GP, then AM of a, b, c is
Answer: 11.00
Solution:

Question: If $3, 7, 11, \ldots, 403 = \text{A.P.}_1$ and $2, 5, 8, \ldots., 401 = A.P._2$. Find the sum of common terms of A.P._1 and A.P._2
Options:
(a) 3366
(b) 6699
(c) 9999
(d) 6666
Answer: (b)
Solution:

Question: The value of integral $\int_0^{\frac{\pi}{4}} \frac{x\,dx}{\cos^4 2x + \sin^4 2x} =$

Options:
(a) $\frac{\pi^2}{16\sqrt{2}}$
(b) $\frac{\pi^2}{64}$
(c) $\frac{\pi^2}{32}$
(d) $\frac{\pi^2}{8\sqrt{2}}$
Answer: (a)
Solution:

$I = \frac{\pi^2}{16\sqrt{2}}$

Question: $y = y(x)$ solution of equation

$$\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1, y(0) = 1\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2 = ?$$

Options:

(a) $\log\frac{4}{4+\sqrt{e}}$

(b) $\frac{2}{1+\sqrt{6}}$

(c) $\frac{3}{3-\sqrt{e}}$

(d) $\frac{1}{2-\sqrt{e}}$

Answer: (d)

Solution:

Question: If the system of equations $2x+3y-z=5;\ x+\alpha y+3z=-4;\ 3x-y+\beta z=7$ have many solutions, then $13\alpha\beta$ is equal to

Options:

(a) 1110

(b) 1120

(c) 1210

(d) 1220

Answer: (b)

Solution:

Question: A bag contains 8 balls, whose colours are either write or black back. 4 balls are drawn at random without replacement and it was found that 2 balls are white and other 2 balls are black. The probability that the bag contains equal number of white and black balls is:

Options:

(a) $\frac{1}{5}$

(b) $\frac{1}{7}$

(c) $\frac{2}{5}$

(d) $\frac{2}{7}$

Answer: (d)

Solution:

Question: For $0 < \theta < \dfrac{\pi}{2}$, if the eccentricity of hyperbola $x^2 - y^2\text{cosec}^2\theta = 5$ is $\sqrt{7}$ times eccentricity of the ellipse $x^2\text{cosec}^2\theta + y^2 = 5$, then the value of θ is:

Options:

(a) $\dfrac{\pi}{6}$

(b) $\dfrac{5\pi}{12}$

(c) $\dfrac{\pi}{4}$

(d) $\dfrac{\pi}{3}$

Answer: (d)

Solution:

Question: Let $s = \left\{ x \in R : \left(\sqrt{3}+\sqrt{2}\right)^x + \left(\sqrt{3}-\sqrt{2}\right)^x = 10 \right\}$. Number of elements in s is:

Options:

(a) 2

(b) 0

(c) 1

(d) 4

Answer: (a)

Solution:

Question: If $A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, C = ABA^T$ and $X = AC^2A^T$ then $|X|$ is equal to

Options:

(a) 729

(b) 283

(c) 27

(d) 23

Answer: (a)

Solution:

Question: $\bar{a} = -5\hat{i} + \hat{j} - 3\hat{k},\ \bar{b} = \hat{i} + 2\hat{j} - 4\hat{k},\ \bar{c} = \left[\left(\bar{a}\times\bar{b}\right)\times\hat{j}\times\hat{j}\right]\times\hat{j}$ then $\bar{c}\cdot\left(-\hat{i}+\hat{j}+\hat{k}\right)$

Answer: -12.00

Solution:

Question: $\dfrac{x-\lambda}{-2} = \dfrac{y-2}{1} = \dfrac{z-1}{1}$ and $\dfrac{x-\sqrt{3}}{1} = \dfrac{y-1}{-2} = \dfrac{z-2}{1}$. If the shortest distance between the above two lines is 1 then sum of possible values of λ

Options:

(a) 0

(b) $2\sqrt{3}$
(c) $3\sqrt{3}$
(d) $-2\sqrt{3}$
Answer: (b)
Solution:

Question: $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2}\cos x}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)} = a\pi + b\log\left(3+2\sqrt{2}\right)$, then find $a+b$.
Options:
(a) 4
(b) 6
(c) 8
(d) 2
Answer: (d)
Solution:

Question: $x = x(t)$ solution of $(t+1)dx = \left[ex + (t+1)^4\right]dx$, $x(0) = 2$ then $x(1) =$
Answer: 12.00
Solution:

Question: Given: $5f(x) + 4f\left(\frac{1}{x}\right) = x^2 - 4$ & $y = 9f(x)\cdot x^2$. If y is strictly increasing, then find interval of x.
Options:
(a) $\left(-\infty, -\frac{1}{\sqrt{5}}\right] \cup \left(\frac{1}{\sqrt{3}}, 0\right)$
(b) $\left(-\frac{1}{\sqrt{5}}, 0\right) \cup \left(0, \frac{1}{\sqrt{5}}\right)$
(c) $\left(0, \frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right)$
(d) $\left(-\sqrt{\frac{2}{5}}, 0\right) \cup \left(\sqrt{\frac{2}{5}}, \infty\right)$
Answer: (d)
Solution: